

COMP 3311: Database Management Systems

Lab 10 Exercise: Oracle Indexing and PL/SQL Exceptions

WHAT TO DO

1. **Download** the zipped folder **Lab10Exercise.zip** from the **Oracle Indexing and PL/SQL Exceptions** entry of the Lab Schedule course webpage and unzip it. The folder contains two script files, **lab10db.sql** and **lab10Queries.sql**, and one text file, **lab10CgaCalculations.txt**.
2. **Place** your **insertMyself.sql** script file inside the **Lab10Exercise** folder.
3. **Modify** your **insertMyself.sql** script file so that it creates a unique index on the attribute **email** of the **Student** table.
4. **Execute** the **lab10db.sql** script file in **Oracle SQL Developer** to confirm that there are no errors in your **insertMyself.sql** script file.
5. **Create** a stored procedure named **lab10DuplicateEmailCheck** in **SQL Developer** that checks whether a unique index exists on the **email** attribute of the **Student** table. To perform this check, the procedure should
 - a. attempt to insert a **Student** record with a known duplicate email (e.g., your student record) and
 - b. handle this implicit exception by outputting the message:
Tried to insert duplicate email into the Student table.

Note: You should use the predefined exception **DUP_VAL_ON_INDEX** that raises an exception when a duplicate value is inserted into a unique index (see the lab notes).
6. **Create** a stored procedure named **lab10CgaCalculations** in **SQL Developer**.
7. **Replace** all the code in your **lab10CgaCalculations** procedure with the code in the file **lab10CgaCalculations.txt**.
8. **Complete** the **TODOs** in the **lab10CgaCalculations** procedure to do the following additional tasks.
 - a. Define a user exception that is raised if the student is an honours student (i.e., the student's cga is greater than or equal to 3.5) and outputs the following message when the exception is raised:
>>> <student name> (<student id>) with CGA=<studentCga> is an honours student.
 - b. If a student has not enrolled in any course, then he/she will not have any related **EnrollsIn** tuples. In this case, when the student's CGA is calculated, the denominator will be zero and a divide-by-zero error will be raised. Therefore, add an exception handler that handles a divide-by-zero error when calculating a student's CGA.

Note: You must use the predefined exception **ZERO_DIVIDE** to check for this error. If a divide-by-zero error occurs, then the exception handler should output the following message:
<student name> (<student id>) is not enrolled in any course.

You should also ensure that if a divide-by-zero error occurs, then *the student's cga is not updated* (i.e., it remains set to null). **Note:** This exception should be handled in a sub-block.
 - c. If the cga is less than or equal to 2, then output the following message:
*** Low CGA alert for <student name> (<student id>) with cga <studentCga>.

In the preceding messages, you should replace <student name> with the concatenation of the first and last name of the student, <student id> with the student's student id and <studentCga> with the student's calculated cga.
9. **After your lab10DuplicateEmailCheck and lab10CgaCalculations procedures execute correctly, execute** the **lab10Queries.sql** script file in **SQL Developer**.

WHAT TO SUBMIT

1. Your completed **lab10DuplicateEmailCheck** procedure code as a text file. You can copy and paste your code from **Oracle SQL Developer** into a text (txt) file.
2. A screenshot of the **Oracle SQL Developer** window or a text file that shows the result of running the **lab10Queries.sql** script file in the Script Output pane as shown in Figure 1.

This is an individual exercise. Each student must submit their own exercise files.

How To SUBMIT

By 6 p.m. tomorrow, upload your completed **lab10DuplicateEmailCheck.txt** text file and the screenshot or text file showing the result of running the **lab10Queries.sql** script file to Canvas by selecting **Lab 10** in the **Lab Assignments** section of the **Assignments** tab on Canvas, and then selecting the **Submit Assignment** button.

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### Tried to insert duplicate email into the Student table.

*** Low CGA alert for David Decker (14567890) with cga 1.49.
### Lazy Lazy (99987654) is not enrolled in any course.
*** Low CGA alert for Donald Trump (26184444) with cga 1.49.
*** Low CGA alert for David Decker (28435018) with cga 1.49.
*** Low CGA alert for Ferris Bueller (66666666) with cga 1.54.
>>> Steve Jobs (15000655) with CGA=3.56 is an honours student.
>>> Alan Turing (28918856) with CGA=3.56 is an honours student.
### Buzzy Bizy (27182481) is not enrolled in any course.
>>> Typical Student (11111111) with CGA=3.56 is an honours student.

```

STUDENTI	FIRSTNAME	LASTNAME	UNIT	CGA
11111111	Typical	Student	COMP	3.56
15000655	Steve	Jobs	COMP	3.56
28918856	Alan	Turing	MATH	3.56
26186666	Warren	Buffet	MGMT	3.42
15085942	Bill	Gates	COMP	3.4
29873381	Nikola	Tesla	ELEC	3.37
13556789	Leo	Greenleaf	MATH	3.36
13782973	Edith	Clarke	ELEC	3.15
28834512	Isaac	Newton	MATH	2.98
15678901	Albert	Einstein	COMP	2.97
13456789	Ariana	Grande	COMP	2.82
13455789	Harry	Potter	COMP	2.76
18792018	Elon	Musk	MGMT	2.76
15678989	Edith	Clarke	COMP	2.73
15456789	Leo	Da Vinci	COMP	2.72
16789012	Robert	Redford	MATH	2.57
26184624	Bruce	Wayne	ELEC	2.47
66666666	Ferris	Bueller	MGMT	1.54
28435018	David	Decker	MGMT	1.49
26184444	Donald	Trump	MGMT	1.49
14567890	David	Decker	ELEC	1.49
99987654	Lazy	Lazy	COMP	
27182481	Buzzy	Bizy	MGMT	

STUDENTI	FIRSTNAME	LASTNAME	UNIT	CGA
66666666	Ferris	Bueller	MGMT	1.54
28435018	David	Decker	MGMT	1.49
14567890	David	Decker	ELEC	1.49
26184444	Donald	Trump	MGMT	1.49

**Your
information
should be
shown in
these lists.**

Figure 1: Example Script Output pane showing the result of executing the **lab10Queries.sql** script file.